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CASE STUDY : 2

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Security the University Examination System

A university uses an operating system - based examin-tion management system to store students records examination marks, faculty details & question papers. the has 3 main user groups. Students, faculty & Administrators. Students should be able to view only their own marks & download premitted learning materials. Faculty members need to view student records & enter or modify marks. Administators manage user accounts, permissions, & system configurt

The university has assigned permissions through roles, but an audit review identifies the following events.

- At 10:00, the acc student 07 has 3 failed login attempts.
- At 10:03, student 07 successfully log in.
- At 10:04, student 04 attempts to open exam marks.xlsx, which contains marks for an entire class.
- At 10:05, the system records an attempted modification of exam marks.xlsx by student 07.
- At 10:15, faculty or modifies marks on Exam marks.xlsx & saves files successfully.
- At 16:20, an administrators discovers that a former faculty member's acc is still active.
- The audit log is stored on the same server & users with broad permission may be able to modify or delete it.

The university wants to protect confidentiality, integrity & availability, prevent unauthorized access, maintain accountability & investigate suspicious

activity. You are asked to reccommend appropriate operating system access control, security - administration & auditing measures.

Questions :-

Q1) Identify the main security problems in the case study. (list atleast 4 risk or weakness indicated by the scenario).

Ans Main security Problems in the Case Study :-

a. Multiple Failed login Attempts :- The students acc has 3 failed login attempts, which may indicate a password - guessing attack.

b. Unauthorized Access :- A student attempts to access exammarks.xlsx, which contains the marks of the entire class.

c. Unauthorized Modification :- The students attempts to modify the examination marks file, threatening the integrity of student records.

d. Inactive Account Not Disabled :- A former faculty member's account is still active creating a risk of unauthorized access.

e. Weak Audit log :- The audit log is stored on the same server, & users with board permissions may modify or delete it, making it difficult to investigate suspicious activities.

Q2) Explain the difference between authentication & authorization using students OT as an example ? (4marks)

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Ans

Authentication	Authorization
• It is the process of verifying the identity of a user. It checks who the user is.	• It is the process of deciding what a user is allowed to access or do. It checks what permissions the user has.
• Eg:- Student 07 enters a valid username & password to log in to the examination sys.	Eg:- Student 07 can view their own marks but cannot modify marks or view the entire class's marks.
• It takes place first	• It takes place after authentication

Q3) Recommand an access control policy for students, faculty & Administrators. Justify your choice.

Ans RBAC for the university examination sys because it gives access according to each user's role.

- Students :- Can view their own marks but cannot change them. Eg:- Student 07 can check only their own result.
- Faculty :- Can enter or update marks of their assigned students. Eg:- A teacher can update marks only for their class.
- Administrators :- Can manage student records, user accounts & permission. Eg:- An admin can create or remove user accounts.
- Security :- It prevents unauthorized people from viewing or changing exam records.

- Monitoring :- The system tracks failed login attempts & changes made to marks.

Conclusion :- RBAC keeps exam records safe by giving each user only the access they need

Q4) Prepare suitable access control list [ACL] for Exammark.xlsx [5 Marks]

Ans An Access Control List [ACL] for exammarks.xlsx defines who can access the file & what actions they can perform.

- Students :- Can view their own marks but cannot edit them. For Eg:- Student 07 can check their result
- Faculty :- Can view & update marks of their assigned students.
- Administrator :- Have full control to view, edit & manage the file.
- Unauthorized users :- Cannot access or modify the file.
- Security monitoring :- Login attempts & changes to marks should be recorded to detect suspicious activity.

This ACL protects examination records from unauthorized access & ensures that each user gets only the permission required for their role.

Q5) Explain how the principle of user privilege should be applied in the sys (4M)

Ans The principle of Least Privilege means giving each user only the access they need to do

their work.

- Student :- Students can check their own marks but cannot change them or see other students's marks.
- Faculty :- Teachers can enter or update marks only for the students assigned to them.
- Administrators :- Admins can manage student records & user accounts as required.
- Removing Access :- If a teacher leaves the university, their accounts should be deactivated so they cannot access the system.

This principle helps keep exam records safe & prevents people from misusing the system.

Q6) Interpret the audit events from 10:00 to 10:05. what should the security team to do next?

Ans The audit events from 10:00 to 10:05 show suspicious activity in the university examination system.

- 10:00 - Failed logins: Student 07 failed to login 3 times, which may indicate an attempt to guess the password
- 10:03 - Successful login: Student 07 logged in successfully after the failed attempt
- 10:04 - Unauthorized Access: Student 07 tried to open the complete exammarks.xlsx file which contains student's marks
- 10:05 - Attempts to Modify marks: student 07 tried to change the examination marks without permissions.

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Next step: The security team should investigate student 07's account, block unauthorized access, check whether any marks were changed, & protect the audit logs from being deleted or modified.
These events indicate a possible security threat so the team should take immediate action to protect the examination data.

Q7) Identify at least 5 events that should be recorded in the audit logs:

Ans Events to be Recorded in the Audit log

- Login Attempts: Successful & failed login attempts, especially those involving student 07.
- File Access: when user opens or tries to access exam marks.xlsx
- Changes to Marks: Any marks added, edited or deleted.
- Permission changes: Any changes in user's access permissions
- Account Activities: Creation, deletion or deactivation of student & faculty acc.

These events help the security team identify suspicious activities & keep examination records safe.

Q8) Explain why the audit logs must be protected & recommended two ways to protect them

Ans Importance & Protection of Audit Logs

Audit logs must be protected because they record login attempts, file access & changes made to examination marks. If someone

changes or deletes these logs, it becomes difficult to investigate security issues.

- Two way to protect audit logs.
  - → Restrict Access: Only authorized administrator should be allowed to access or manage audit logs.
  - → Secure Backup: Store copies of audit logs on a separate, secure server to prevent them from being easily deleted or modified.

Q9) What action should to be taken regarding the former faculty member's office acc?

Ans Action for former Faculty Member's Account

The former faculty member's account should be deactivated immediately because they no longer work at the university.

- Deactivate Account: Disable the account so they cannot log in.
- Remove Access: Remove their access to files like exammarks.xlsx & other university data
- Check Account Activity: Check their recent activities to see if they accessed anything without permission.

This helps prevent unauthorized access & keeps the university's data safe.

Q10) Suggest a practical incident-response & corrective-action plan for the university?

Ans Incident-Response & corrective Action Plan

- Investigation : Check the audit logs to find out what student 07 did & whether any marks in exammarks. x|sx were changed.
- Access Review : Deactive the former faculty member's account & remove unnecessary access from other users.
- Policy changes : Set strong password rules, give access according to user's roles & protect audit logs from being changed or deleted.
- Follow-up monitoring : Regularly check login attempts & changes made to examination marks to identify suspicious activities.

These steps will help the university protect examination records & prevent similar security problems in the future.